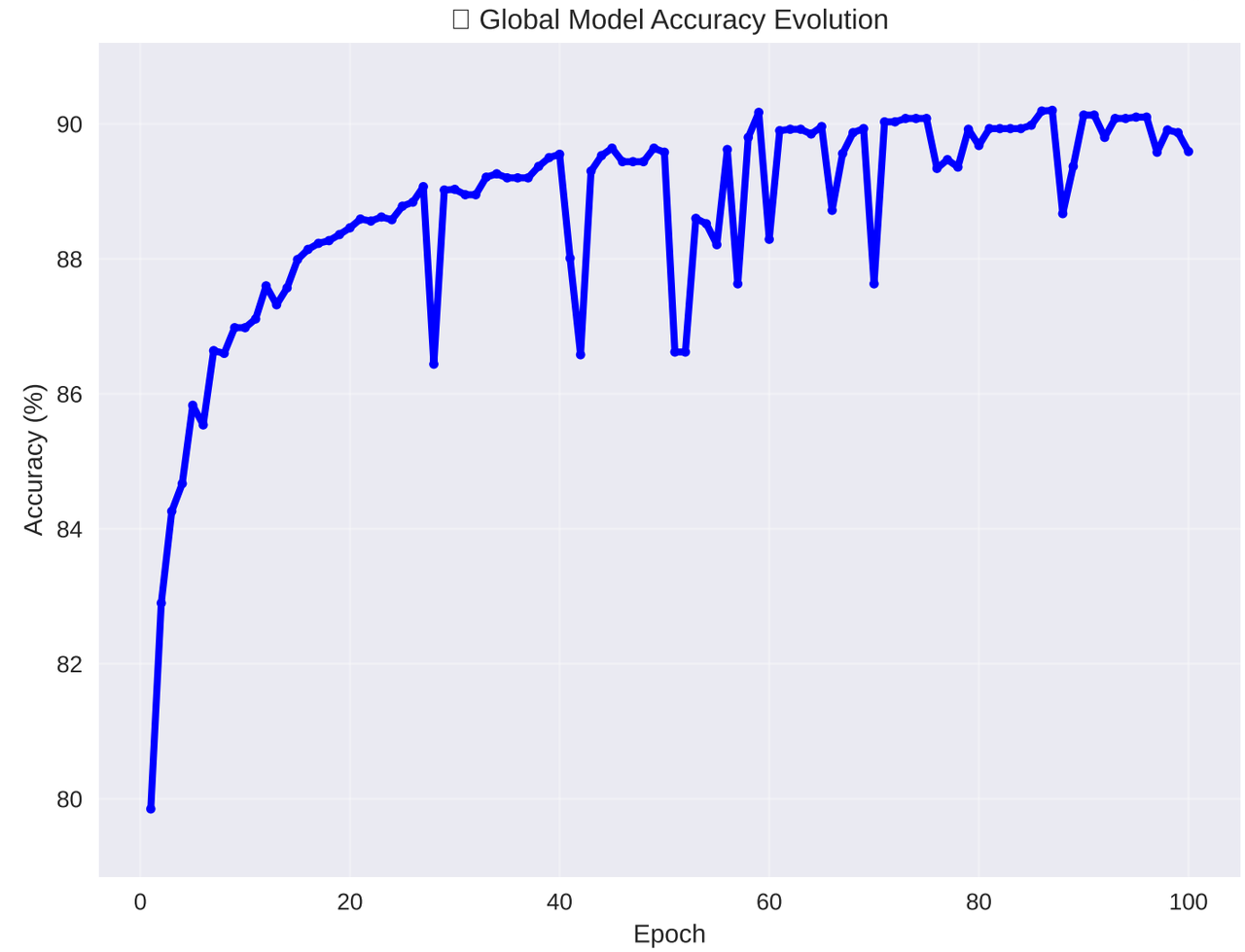
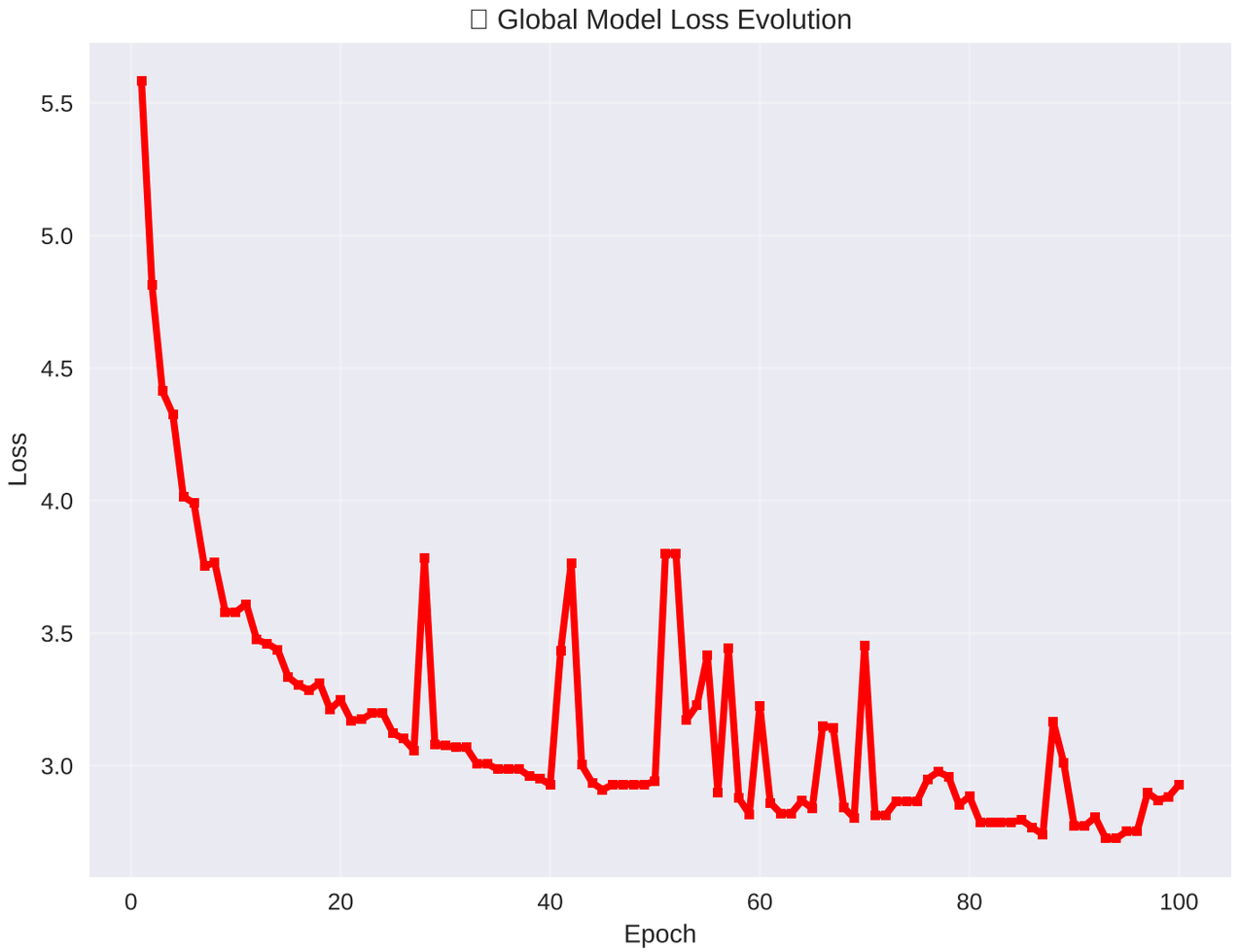
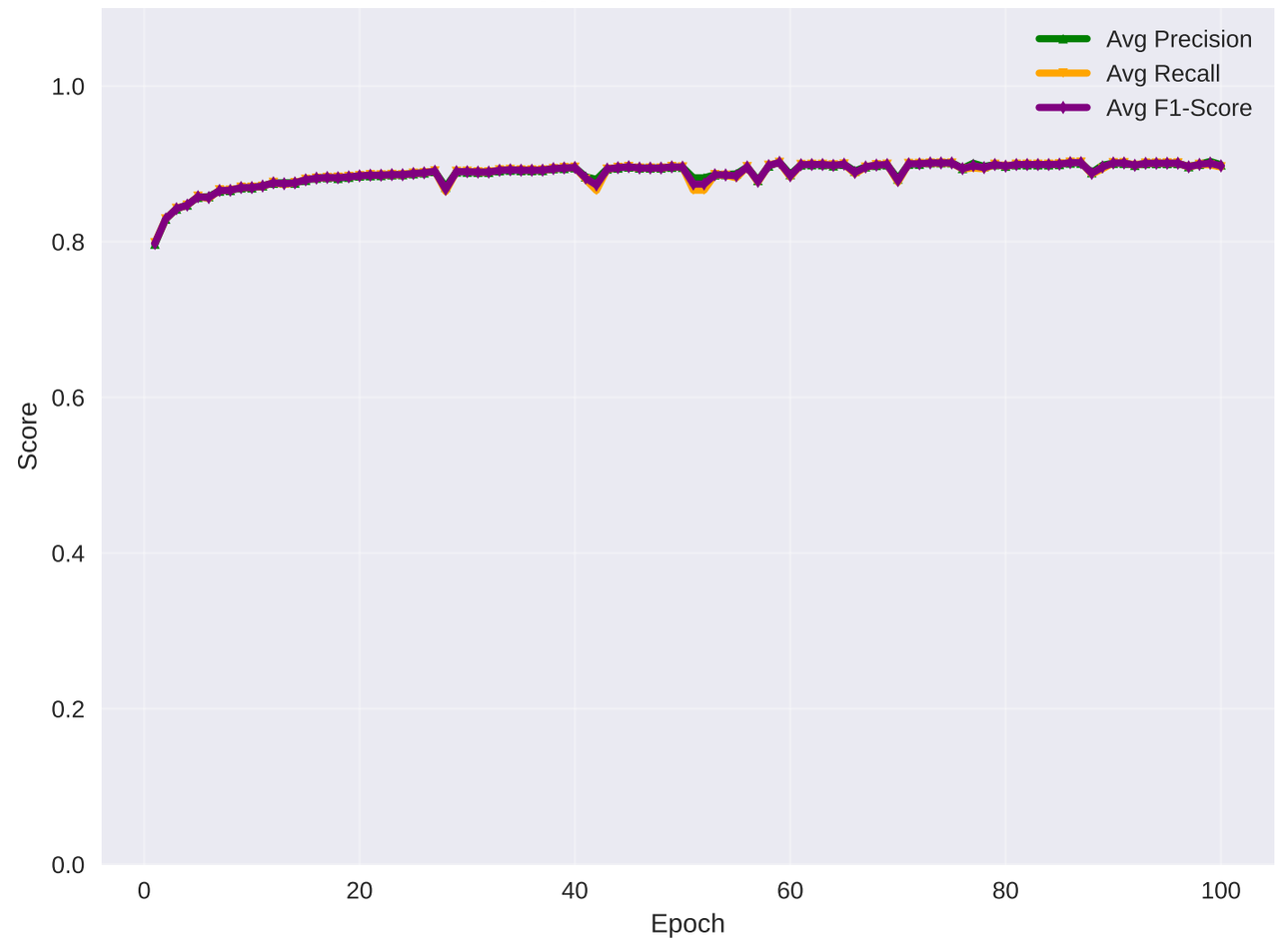
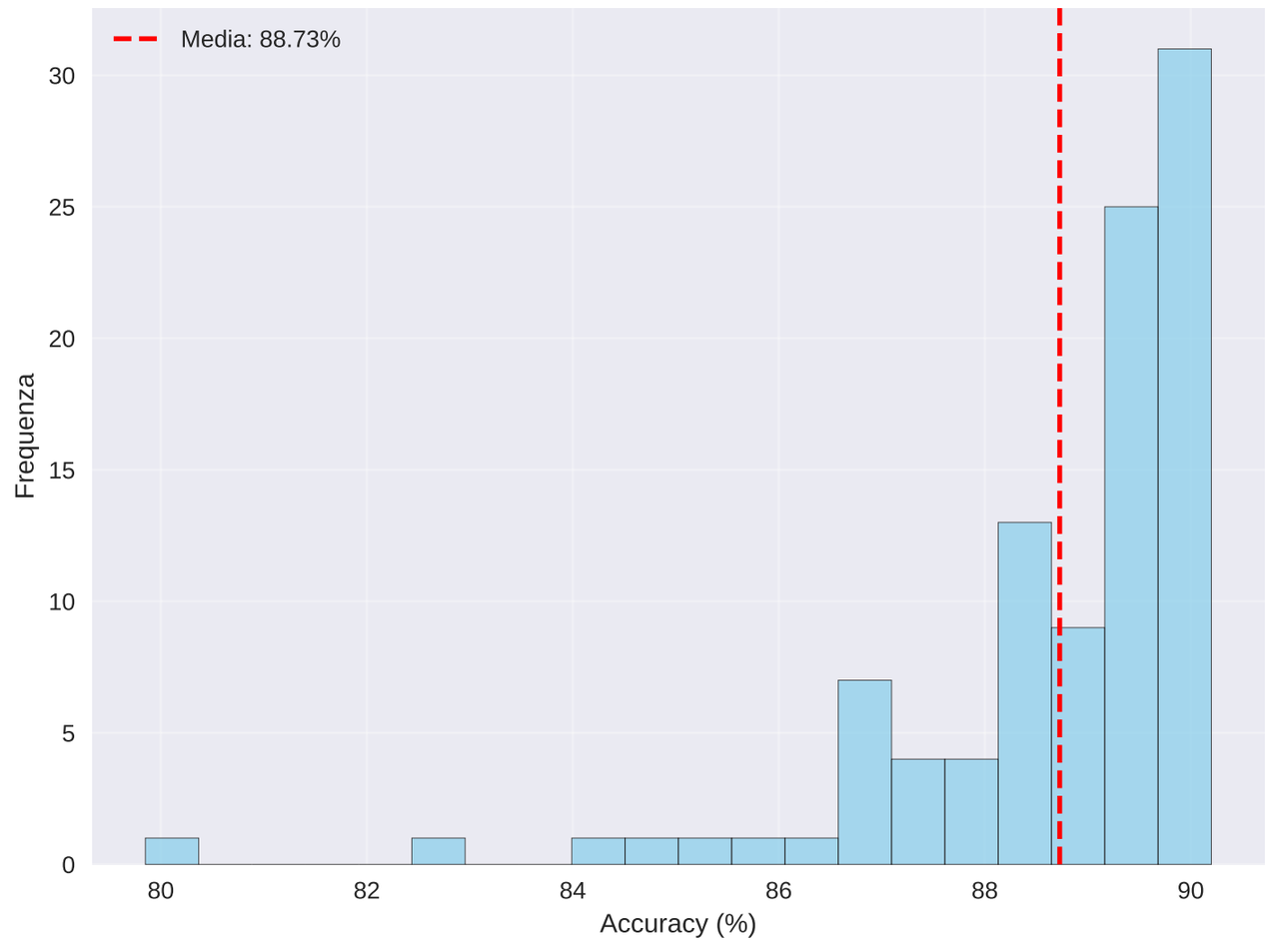
Federated Learning - Main Metrics Overview

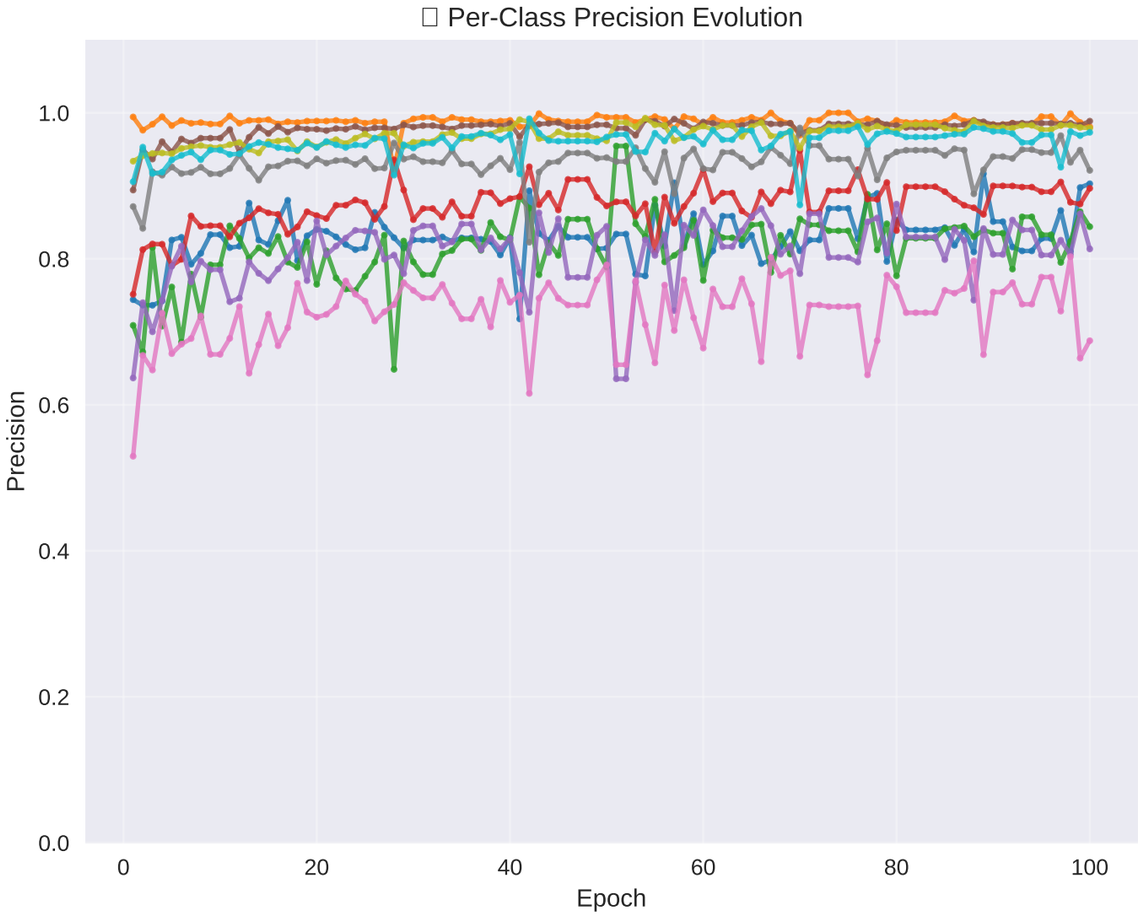


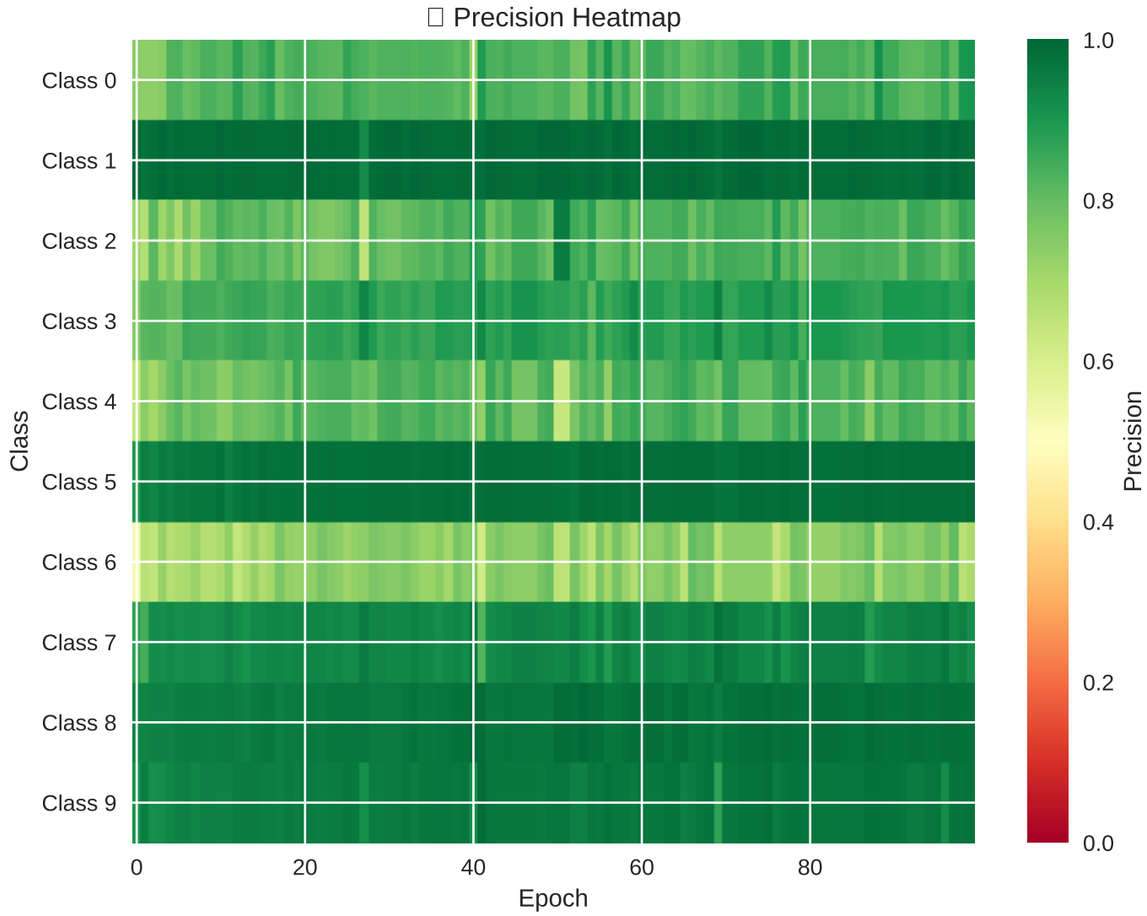


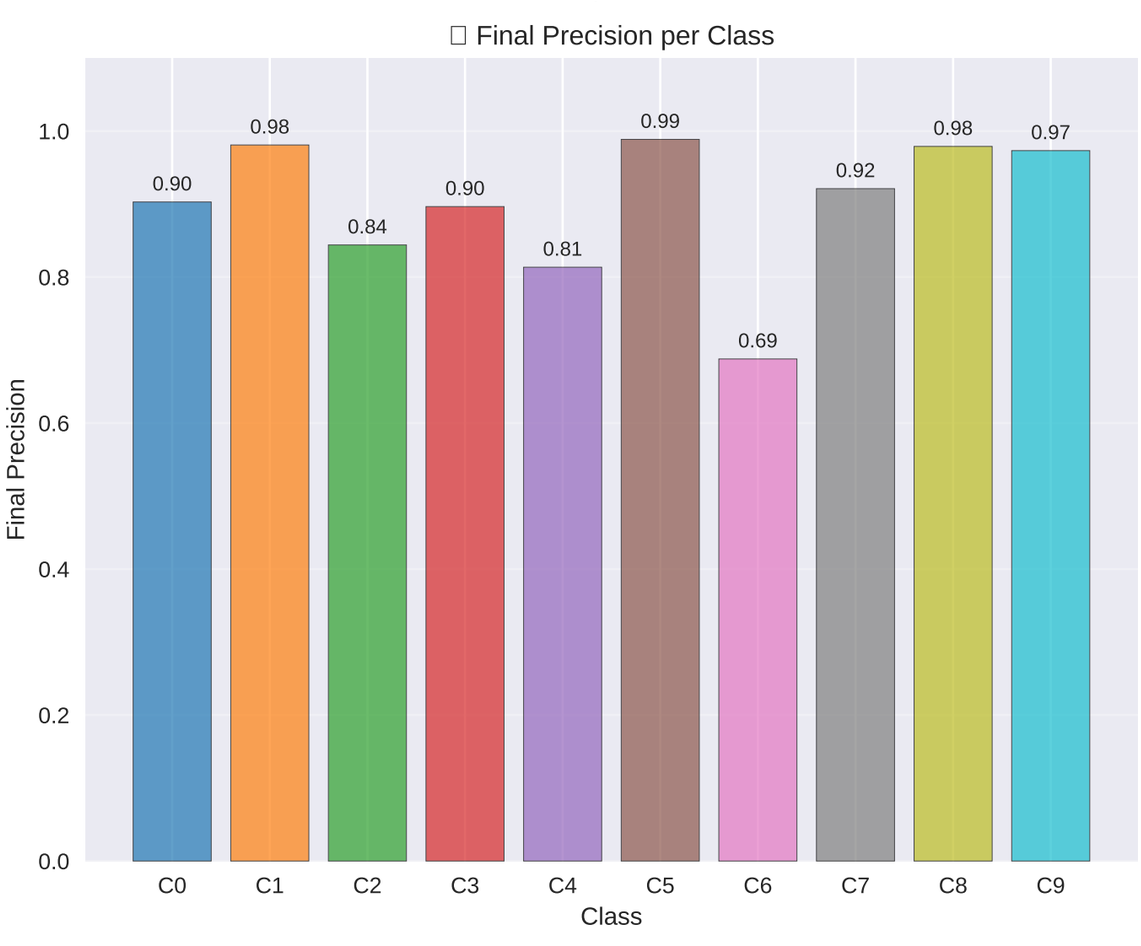


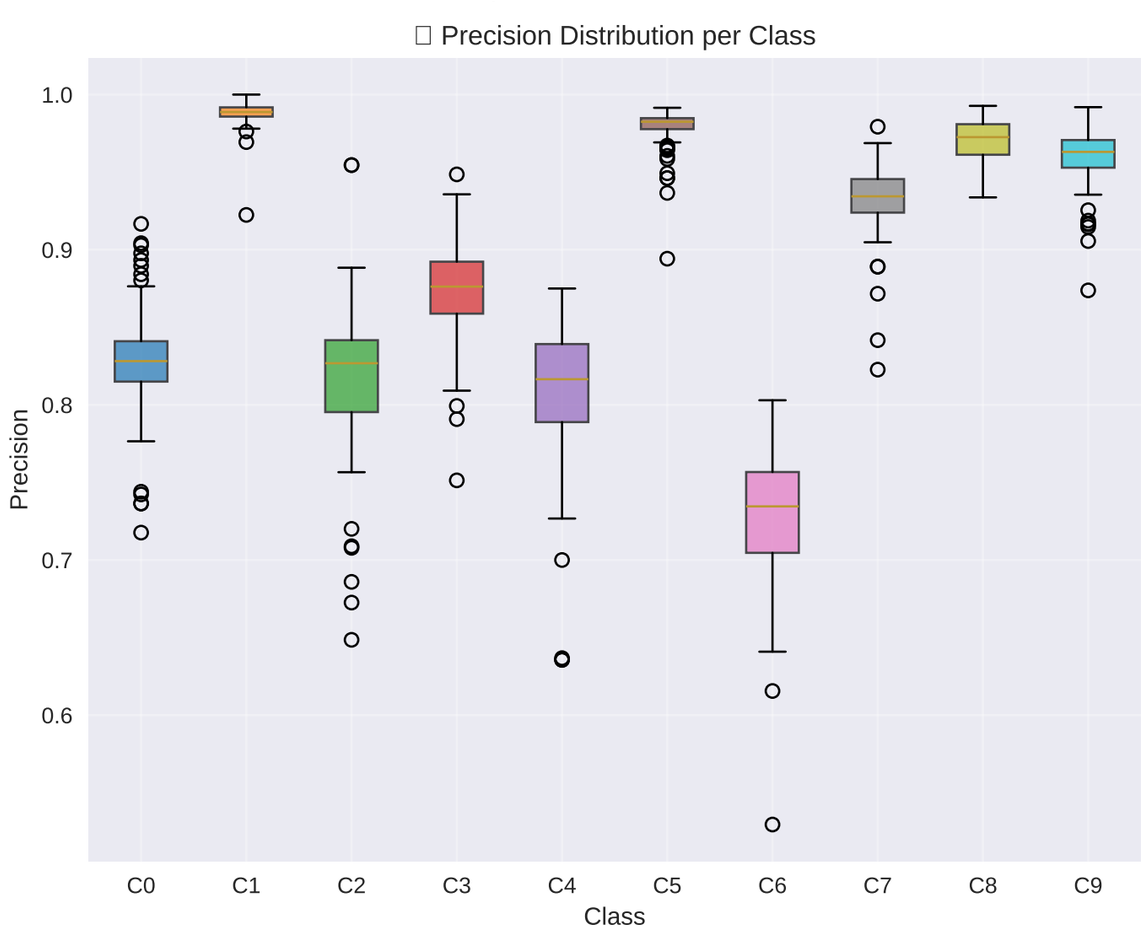


Per-Class Precision Analysis



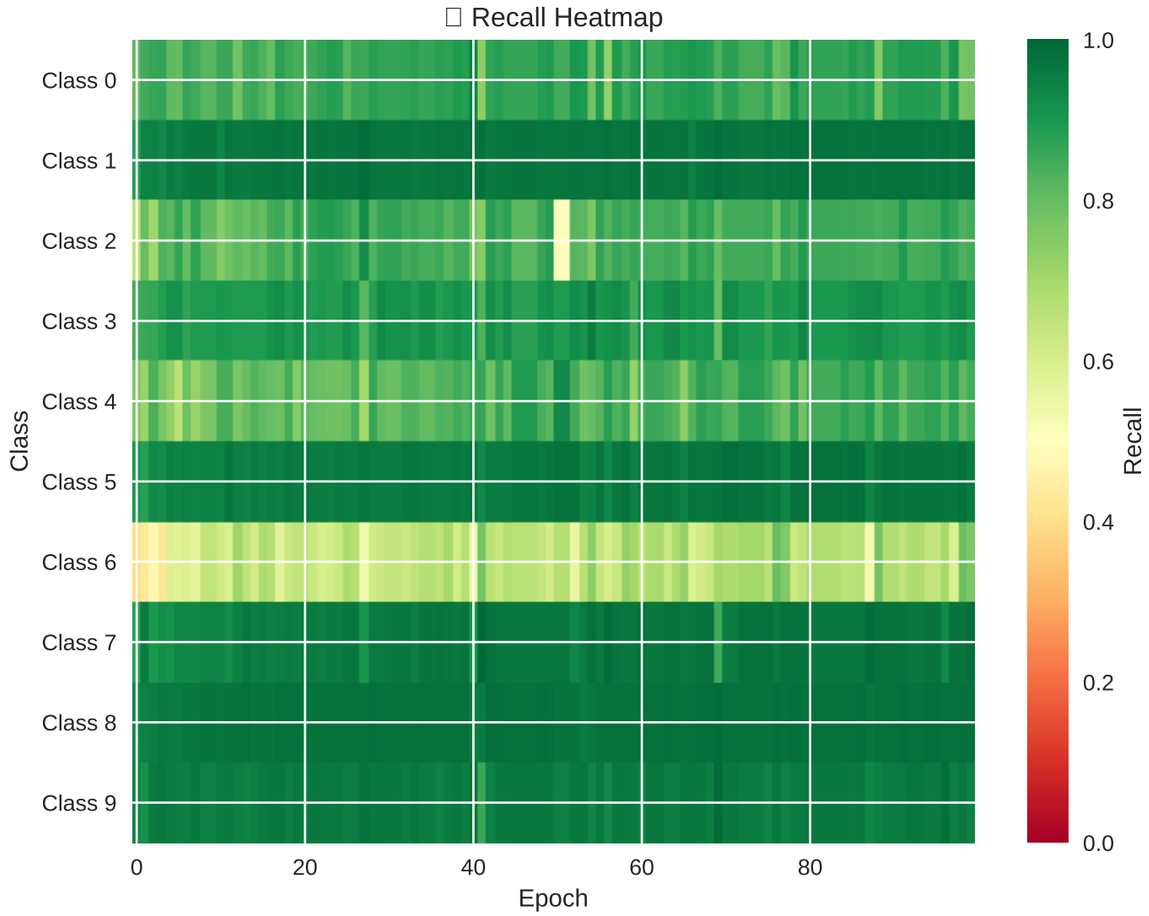


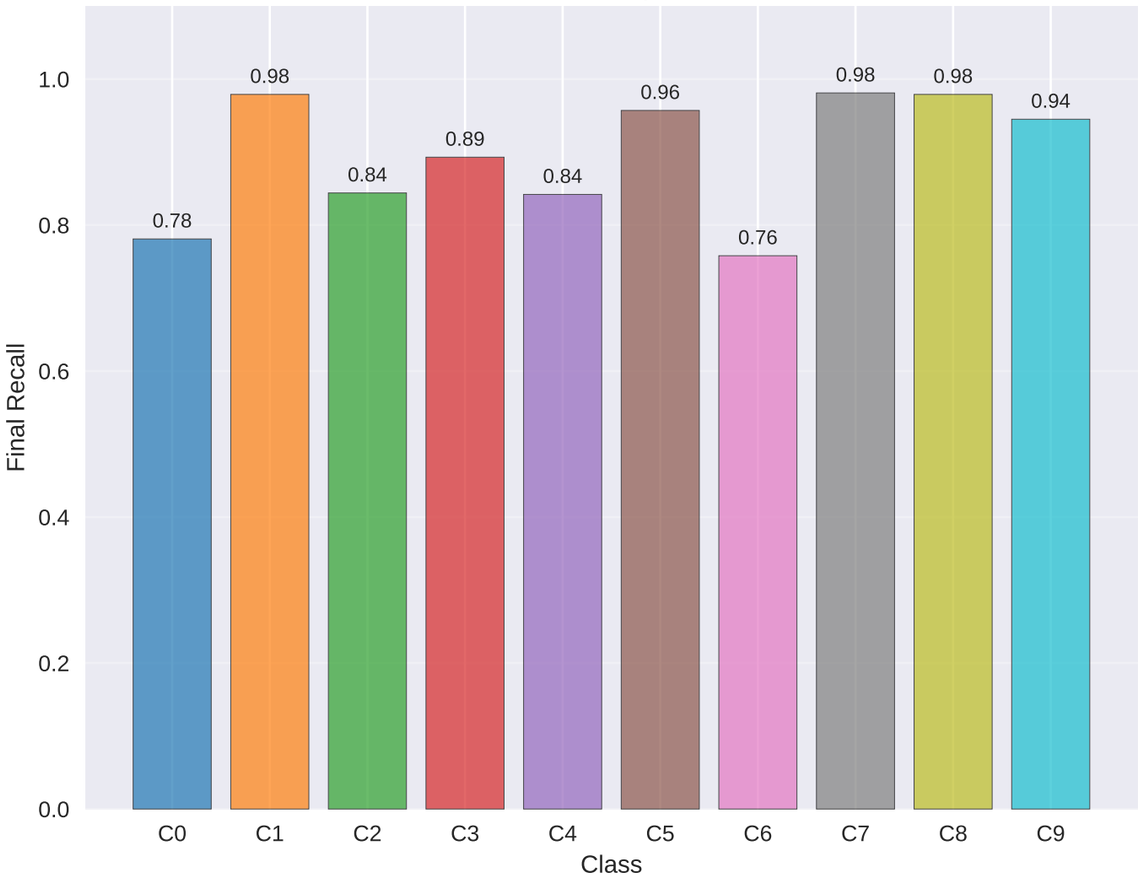


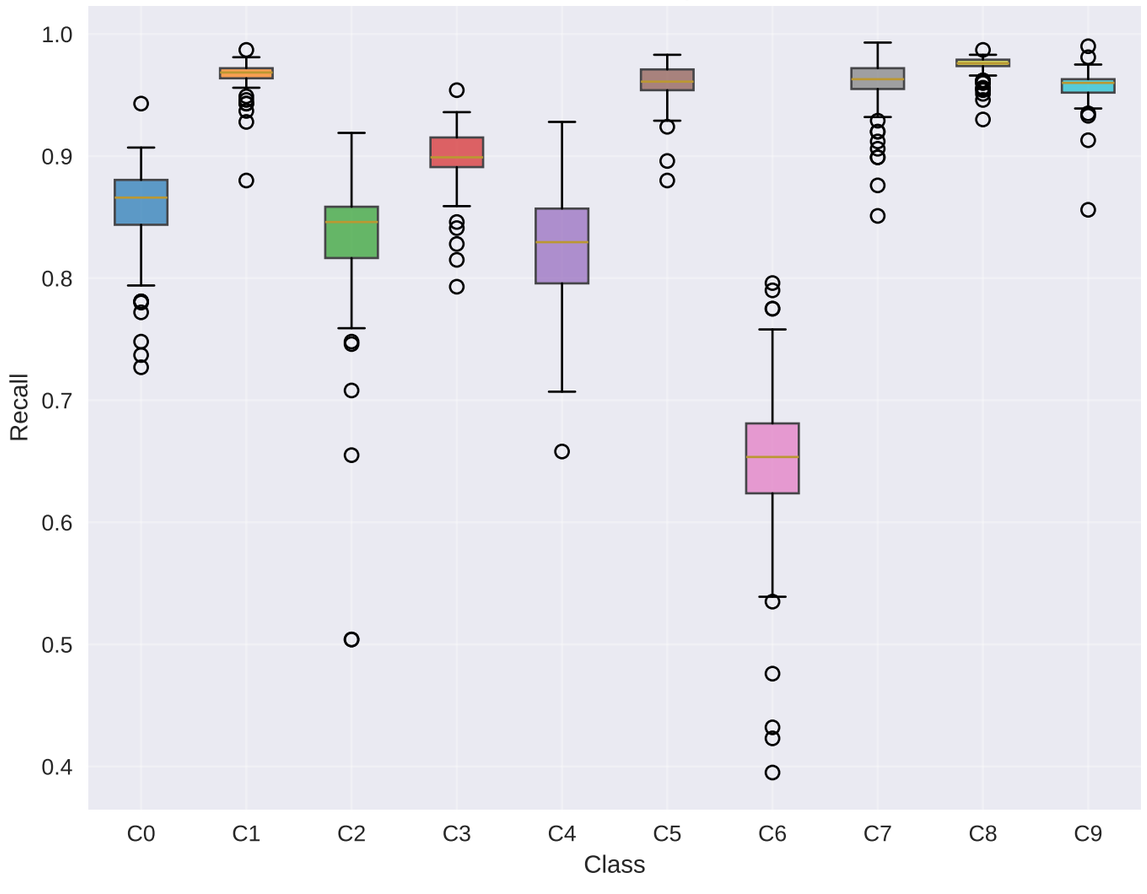


Per-Class Recall Analysis

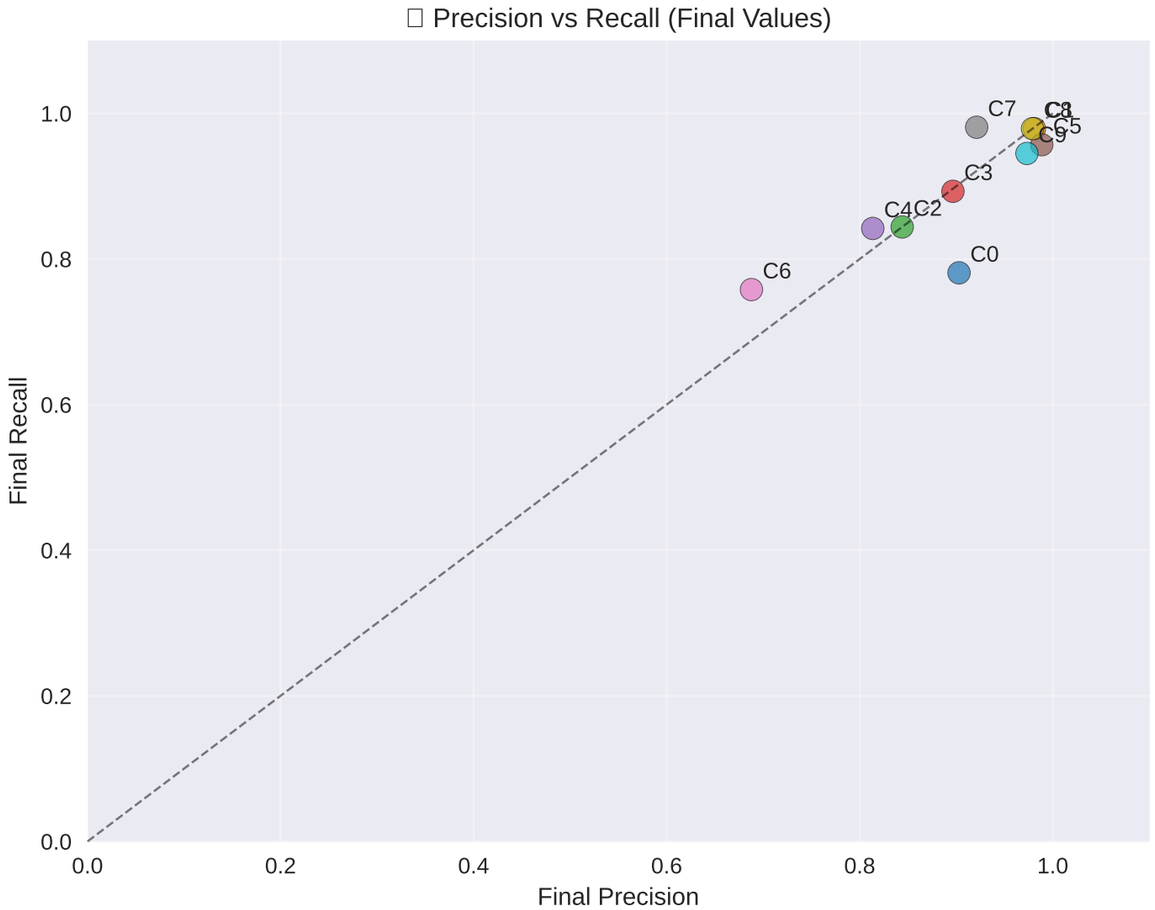




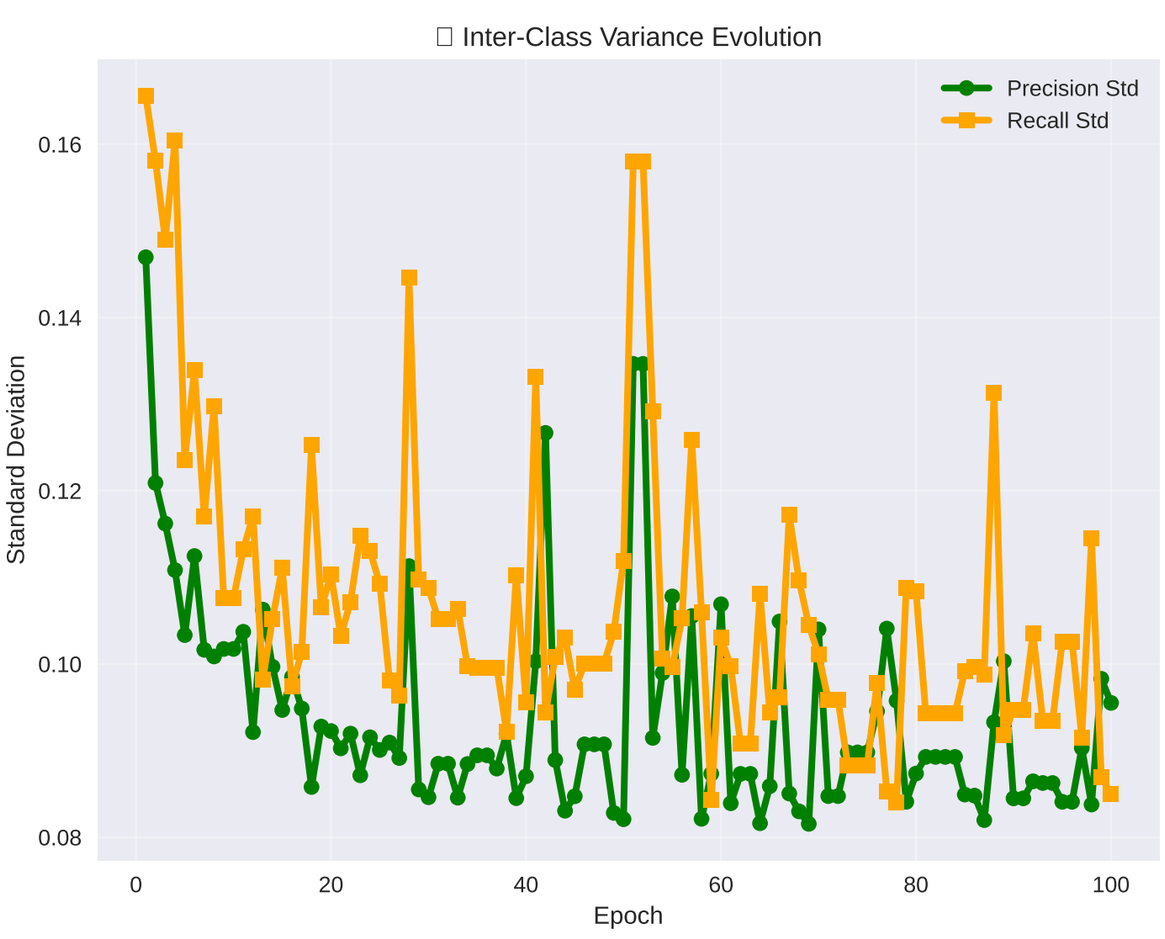


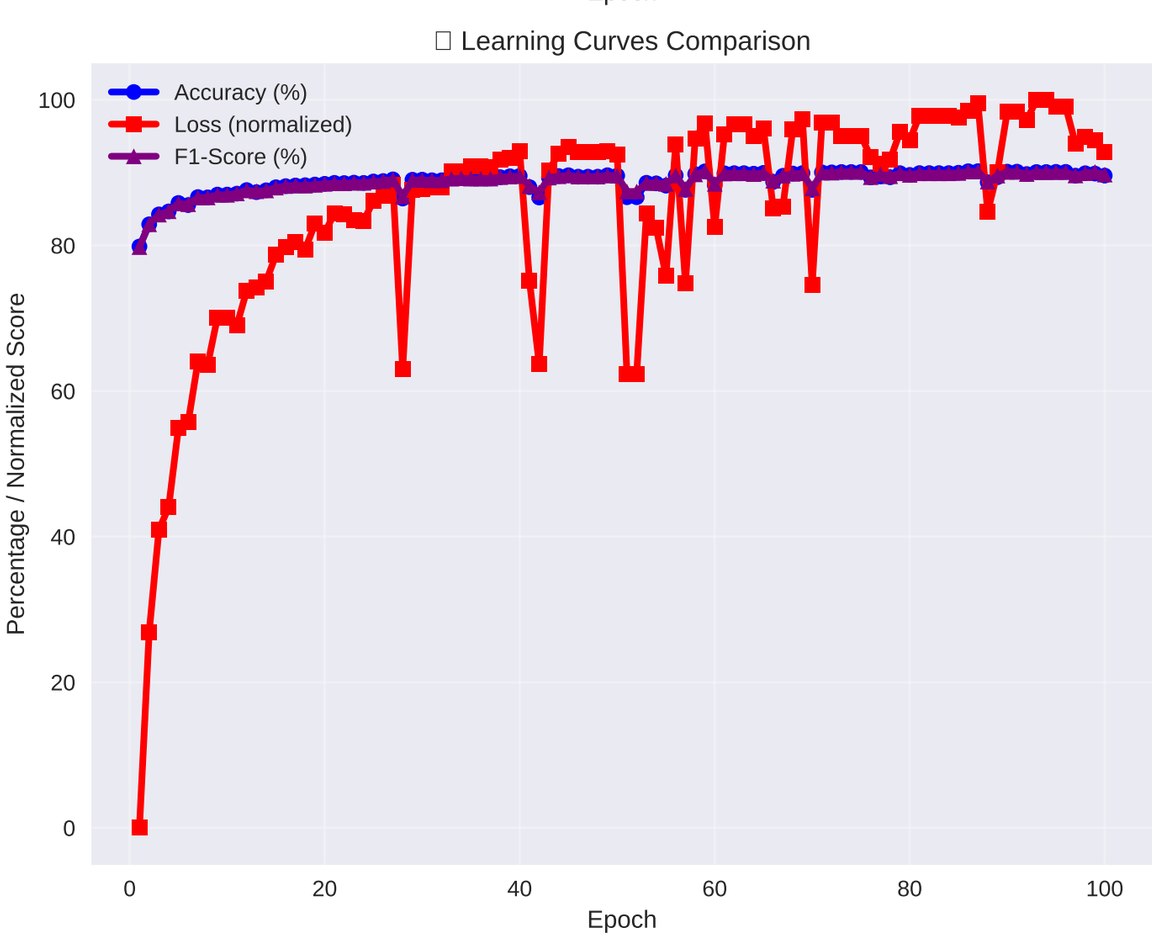


Comparative Analysis









Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-09-10 12:41:07
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 89.59%
• Loss: 2.9290
• Average Precision: 0.899
• Average Recall: 0.896
• Average F1-Score: 0.897

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +9.74%
• Riduzione Loss: +2.6550
• Stabilità Accuracy (CV): 0.0189

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 5 (0.989)
• Peggiore Precision: Classe 6 (0.688)
• Migliore Recall: Classe 7 (0.981)
• Peggiore Recall: Classe 6 (0.758)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.903, R=0.781, F1=0.838
• Classe 1: P=0.981, R=0.979, F1=0.980
• Classe 2: P=0.844, R=0.844, F1=0.844
• Classe 3: P=0.897, R=0.893, F1=0.895
• Classe 4: P=0.814, R=0.842, F1=0.828
• Classe 5: P=0.989, R=0.957, F1=0.973
• Classe 6: P=0.688, R=0.758, F1=0.721
• Classe 7: P=0.921, R=0.981, F1=0.950
• Classe 8: P=0.979, R=0.979, F1=0.979
• Classe 9: P=0.973, R=0.945, F1=0.959

VALUTAZIONE QUALITATIVA:
• Performance Globale: MEDIA
• Stabilità Training: STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
• Considerare aumento epoche o tuning iperparametri
• Implementare class balancing o weighted loss
```